

Tony Huynh

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EDUCATION

The University of Texas at Austin

Dec. 2027

B.S. in Computer Science, Minor in Data Science and Robotics

Austin, TX

Honors/Awards: 4x University Honors, National Cyber Scholar

Coursework: Data Structures, Operating Systems, Algorithms, Software Engineering, Machine Learning

EXPERIENCE

Field Application Engineer Intern

Feb. 2026 – May 2026, Aug. 2026 – Present

Samsung Austin Semiconductor

Austin, TX

- Leading site robotic operations, delivering technical insights to engineering teams and on-site stakeholders.
- Systematically evaluating **Field Foundation Models (FFMs)** across perception and navigation pipelines.
- Orchestrated shell scripts to automate remote data capture and secure data delivery via **rsync** and **SCP**.

AI & Robotics Engineer Intern

June 2025 - Aug. 2025, June 2026 - Aug. 2026

Bechtel Corporation

Sugar Land, TX

- Trained a neural net to predict site assets from drone photogrammetry, achieving **93%** accuracy on unseen data.
- Engineered an autonomous rover using **Python/Lua/C++**, ArduPilot, and an NVIDIA Jetson Orin, integrating **LiDAR** and **RealSense** cameras for real-time spatial awareness.
- Refactored **2,000+** lines of Python obstacle avoidance logic into C++, boosting refresh rate from **3Hz** to **27Hz**.
- Tuned **PID** controllers using **RTK-GNSS** feedback, achieving centimeter-level autonomous navigation accuracy.
- Built a full-stack app with a **FastAPI** backend, polling **MAVLink**/GPS telemetry for real-time rover monitoring.

Undergraduate Researcher

Jan. 2025 – Dec. 2025

Living with Robots Lab - University of Texas at Austin

Austin, TX

- Built a pose-anchored following pipeline with **YOLOv11** and **DeepSORT**, back-projecting hip keypoints and fusing appearance/motion cues for identity verification, cutting target jitter by **39%** and pause events by **54%**.
- Developed Bookie, a socially compliant library service robot that integrates Meta's **LLaMA 3.2 3B** language model with **ROS2** for real-time navigation and human-robot interaction.
- Engineered a custom message queue that connected **SQL**, an LLM-prompt system, and user interface.

Machine Learning Research Assistant

Sep. 2024 – Apr. 2025

Cambronne Lab - University of Texas at Austin

Austin, TX

- Fine-tuned deep learning models used for protein function predictions using **LoRA**, improving base line models by **15-20%** on a dataset of **4600+** complex protein sequences.
- Automated model benchmarking for tuned models to enable continuous integration and iterative evaluation.

PROJECTS

Spotify Recommender App | *MongoDB, ExpressJS, React, Node.js, FastAPI, Python, Docker*

- Developed a full-stack Spotify clone with **ML** microservices, using **REST** communication between services.
- Implemented a content-based recommender trained on **90,000** tracks across **122-dimensional** features.
- Designed a multi-tiered fallback recommendation system based on user feedback captured in **MongoDB**.

BudgeX | *Electron, React, TypeScript, SQLite*

- A personal finance application featuring local CRUD management and real-time stock price ingestion.
- Developed a **React/TypeScript** frontend powered by an **Electron** backend, integrating **Drizzle ORM**, **SQLite**, and external **APIs**.

TECHNICAL SKILLS

Languages: Java, Python, Lua, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, R

Frameworks: React, Node.js, Express.js, Electron, Material-UI, FastAPI, PyTorch/TensorFlow, ROS2

Developer Tools: Git, Copilot, Claude Code, Docker, Google Cloud Platform, MongoDB, Jupyter, VS Code, Linux

Libraries: Scikit-learn, OpenCV, pandas, NumPy, TensorRT, Matplotlib, Drizzle ORM

EXTRACURRICULAR/CAMPUS INVOLVEMENT

Austin Bodybuilding (ABB), Texas Robotics, Texas ACM, Machine Learning and Data Science (MLDS)